INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Domaine Scientifique de la DOUA
20, Avenue Albert Einstein - 69100 Villeurbanne

Cybersecurity

Cybersecurity

IDENTIFICATION

CODE : GI-5-S1-EC-CYB
ECTS : 2

HOURS

Cours :	0h
TD :	6h
TP :	12h
Projet :	0h
Evaluation :	2h
Face à face pédagogique :	20h
Travail personnel :	20h
Total :	40h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. ZAMAI Eric :
eric.zamai@insa-lyon.fr

AIMS

CONTENT

Objectives: Understand the challenges related to cybersecurity in industrial systems and the specific features of this field. Understand the basic elements for identifying the vulnerabilities of these systems, as well as the recommendations and methodology for strengthening the cybersecurity level of existing systems and the key points for designing new systems. Understand the cybersecurity stakeholders in companies and support the implementation of the approach. Understand how to communicate with IT and OT to unite teams around these new challenges.
Main concepts of cybersecurity for information systems in general:
Definitions of cybersecurity and main concepts
Challenges, current status, history;
Classical attacks (MITM, spoofing, social engineering, denial of service, session hijacking, DDOS, APT, worms);
Specific attacks (malicious organizations): PLC code hijacking, takeover of the SCADA/MES information system, takeover of PLCs and industrial equipment
Duality of operational safety and cybersecurity; Examples of attacks (study, practical work on platform attacks);
Classic vulnerabilities and attack vectors;
Main principles for implementing a cybersecurity project (risk analysis, DEP, PSSI, etc.)
Best practices and recommendations;
Overview of norms and standards (2700X, product certification, etc.);
In France, the LPM;
Introduction to cryptography (symmetric/asymmetric encryption, hash functions, signatures, etc.).
IDS modeling (SED, ECC link)
Configuring industrial firewalls and understanding bad ideas.
The industrial system cybersecurity project in a real environment (transfer room practical work);Vulnerability analysis
Security strengthening
In the shoes of a cyber attacker

BIBLIOGRAPHY

PRE-REQUISITES

Basics of automation and different types of industrial systems;
Composition of an industrial system (ECC);
PLC programming languages (ECC, COP, TP API);
Industrial protocols and fieldbuses (ECC);
Classic network architectures (ECC, COP);
Operational safety (CPS/SDF to be extended with SIL standards and consolidate FMEA + EBIOS);
Overview of norms and standards (ECC, API labs, COP, SCADA/MES)
Computer Basics:
Concepts of classic hardware architectures (microprocessor, memory, bus, peripherals, firmware)
Operating systems (main systems used)
Concepts of software architectures, databases
Network Basics:
Main concepts and vocabulary used in this field:
o 7-layer OSI model,
o Data units (frame, packet, segment, etc.),
o MAC address, IP address, port concept (UDP/TCP), o Network discovery and diagnostics (ARP, ICMP),
o Connected (TCP) and offline (UDP) modes.
Services that may be present in industrial equipment (PLC, MES, ERP, HMI, VFD, wireless terminal, etc.)
o Web (HTTP) and secure Web (HTTPs),
o Equipment management (SNMP, SYSLOG, etc.). VPN using a VPN gateway,
router,
gateway,
firewall.
Information systems fundamentals
Data warehouses and data lakes